

Class 10: Structural Bioinformatics 1

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Background

The main repository of high-resolution structural data on biomolecules is called the **Protein Data Bank** (PDB).

PDB statistics

What is in the PDB in terms of molecule type and structure determination method?

Read a CSV file of current PDB stats obtained from <https://www.rcsb.org/stats/summary>

```
pdb <- read.csv("Data Export Summary.csv")
pdb
```

	Molecular.Type	X.ray	EM	NMR	Integrative	Multiple.methods
1	Protein (only)	180,758	23,111	12,813	348	229
2	Protein/Oligosaccharide	10,488	3,741	34	8	11
3	Protein/NA	9,205	6,751	287	26	8
4	Nucleic acid (only)	3,154	250	1,578	3	15
5	Other	178	27	35	4	0
6	Oligosaccharide (only)	11	0	6	0	1
	Neutron Other Total					
1	84 32	217,375				
2	1 0	14,283				
3	0 0	16,277				
4	3 1	5,004				
5	0 0	244				
6	0 4	22				

Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

```
pdb$X.ray
```

```
[1] "180,758" "10,488" "9,205" "3,154" "178" "11"
```

This print out above `pdb$X.ray` is “character” not “numeric”. Therefore I can’t do math with it. We need to fix this...

Two functions that can help here are `sub()` and `as.numeric()`

```
# We want to get rid (or sub out) commas:  
x <- pdb$X.ray  
tmp <- sub(",", "", x)  
sum( as.numeric(tmp) )
```

```
[1] 203794
```

We could make a function to do this:

```
rm.comma <- function(x) {  
  tmp <- sub(",", "", x)  
  sum( as.numeric(tmp) )  
}
```

```
n.tot <- rm.comma(pdb$Total)  
n.xray <- rm.comma(pdb$X.ray)  
n.em <- rm.comma(pdb$EM)  
  
n.xray/n.tot * 100
```

```
[1] 80.48577
```

```
n.em/n.tot * 100
```

```
[1] 13.38046
```

We could also use a different import function `read_csv()` for this CSV that speaks American (i.e. deals with commas in numbers in a comma separated value file).

```
library(readr)
```

Warning: package 'readr' was built under R version 4.4.3

```
pdb <- read_csv("Data Export Summary.csv")
```

Rows: 6 Columns: 9

-- Column specification -----

Delimiter: ","

chr (1): Molecular Type

dbl (4): Integrative, Multiple methods, Neutron, Other

num (4): X-ray, EM, NMR, Total

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
n.tot <- sum(pdb$Total)
```

```
n.xray <- sum(pdb$`X-ray`)
```

```
n.em <- sum(pdb$EM)
```

```
n.xray / n.tot * 100
```

```
[1] 80.48577
```

```
n.em / n.tot * 100
```

```
[1] 13.38046
```

Q. How many total protein structures are there in the dataset?

```
pdb$Total[1]
```

```
[1] 217375
```

The total number of protein sequences in UniProt is 202,556,314

217375/202556314 * 100

[1] 0.1073158

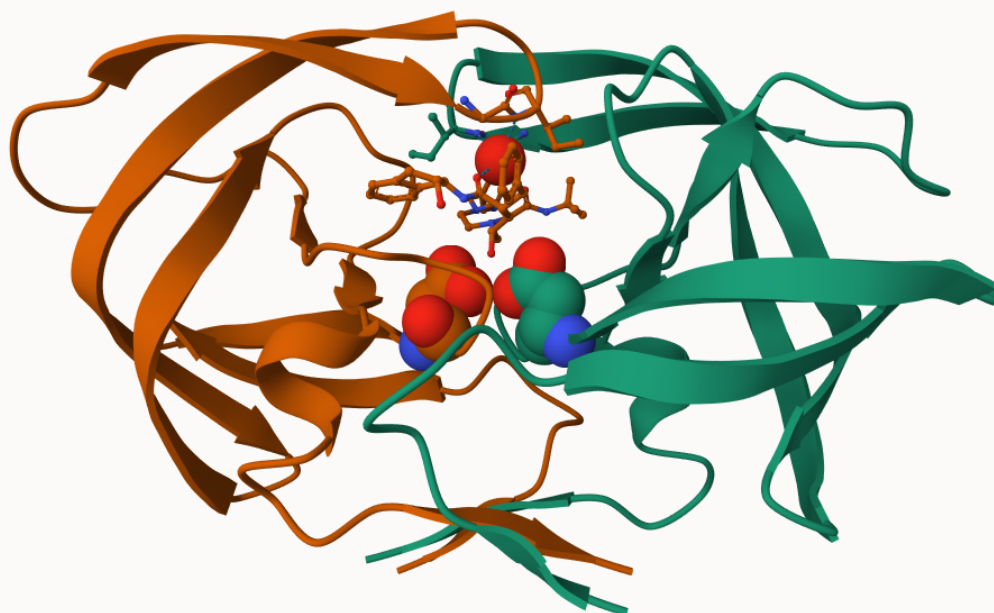
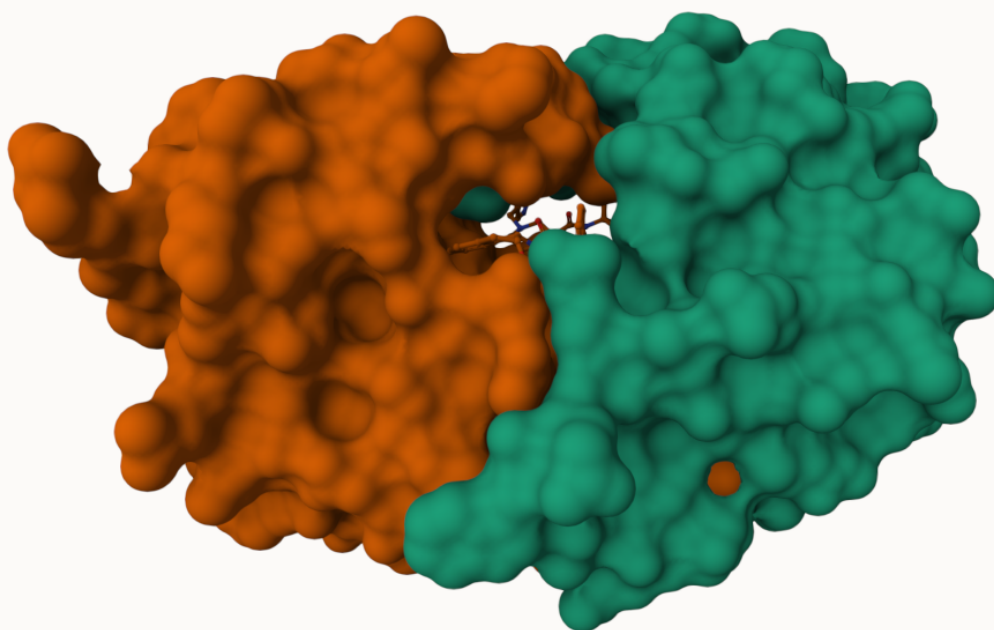
Key-point: We have a very, very small structural coverage of known proteins (~0.1%). Most structures we know about (~80%) come from one method (X-ray crystallography).

Visualizing PDB data with Mol-star

Main stand alone web version with all features is at <https://molstar.org/viewer/>



Figure 1: The HIV-protease enzyme is a homodimer with two chains



```
library(bio3d)  
adk <- read.pdb("6s36")
```

```
Note: Accessing on-line PDB file
PDB has ALT records, taking A only, rm.alt=TRUE
```

```
aa <- get.seq("lake_A")
```

Warning in get.seq("lake_A"): Removing existing file: seqs.fasta

Fetching... Please wait. Done.

```
s <- seqaln(
  seqbind(pdbseq(adk), aa), exefile="msa" )
```

S

```

seq1      1      .      .      .      .      .      60
pdb|1AKE|A MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLRAAVKSGSELGKQAKDIMDAGKLVT
          1      .      .      .      .      .      60
          *****

seq1      61      .      .      .      .      .      120
pdb|1AKE|A DELVIALVKERIAQEDCRNGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVDKI
          61      .      .      .      .      .      120
          *****~*

seq1      121      .      .      .      .      .      180
pdb|1AKE|A VGRRVHAPSGRVYHV KFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQM TAPLIG
          121      .      .      .      .      .      180
          *****

seq1      181      .      .      .      214
pdb|1AKE|A YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
          181      .      .      .      214
          *****

```

Call:

```
seqaln(aln = seqbind(pdbseq(adk), aa), exefile = "msa")
```

Class:

fasta

Alignment dimensions:

2 sequence rows; 214 position columns (214 non-gap, 0 gap)

+ attr: id, ali, call

```
#
```

```
#b <- blast.pdb( pdbseq(adk) )
```

```
#hits <- plot(b)
```

```
#hits$pdb.id
```

```
hits <- NULL
```

```
hits$pdb.id <-c('1AKE_A','6S36_A','6RZE_A','3HPR_A','1E4V_A','5EJE_A','1E4Y_A','3X2S_A','6HA
```

```
files <- get.pdb(hits$pdb.id, path="pdbs", split=TRUE, gzip=TRUE)
```

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1AKE.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6S36.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6RZE.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3HPR.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4V.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/5EJE.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4Y.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3X2S.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAP.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAM.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4K46.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3GMT.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4PZL.pdb.gz exists. Skipping download



	=====	69%
	=====	77%
	=====	85%
	=====	92%
	=====	100%

```
pdbbs <- pdbaln(files, fit = TRUE, exefile="msa")
```

Reading PDB files:

```
pdbbs/split_chain/1AKE_A.pdb
pdbbs/split_chain/6S36_A.pdb
pdbbs/split_chain/6RZE_A.pdb
pdbbs/split_chain/3HPR_A.pdb
pdbbs/split_chain/1E4V_A.pdb
pdbbs/split_chain/5EJE_A.pdb
pdbbs/split_chain/1E4Y_A.pdb
pdbbs/split_chain/3X2S_A.pdb
pdbbs/split_chain/6HAP_A.pdb
pdbbs/split_chain/6HAM_A.pdb
pdbbs/split_chain/4K46_A.pdb
pdbbs/split_chain/3GMT_A.pdb
pdbbs/split_chain/4PZL_A.pdb
  PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..  PDB has ALT records, taking A only, rm.alt=TRUE
.... PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
...
```

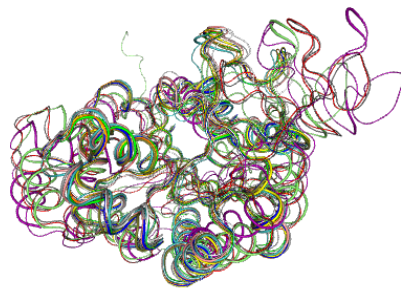
Extracting sequences

```
pdb/seq: 1   name: pdbbs/split_chain/1AKE_A.pdb
  PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 2   name: pdbbs/split_chain/6S36_A.pdb
  PDB has ALT records, taking A only, rm.alt=TRUE
```

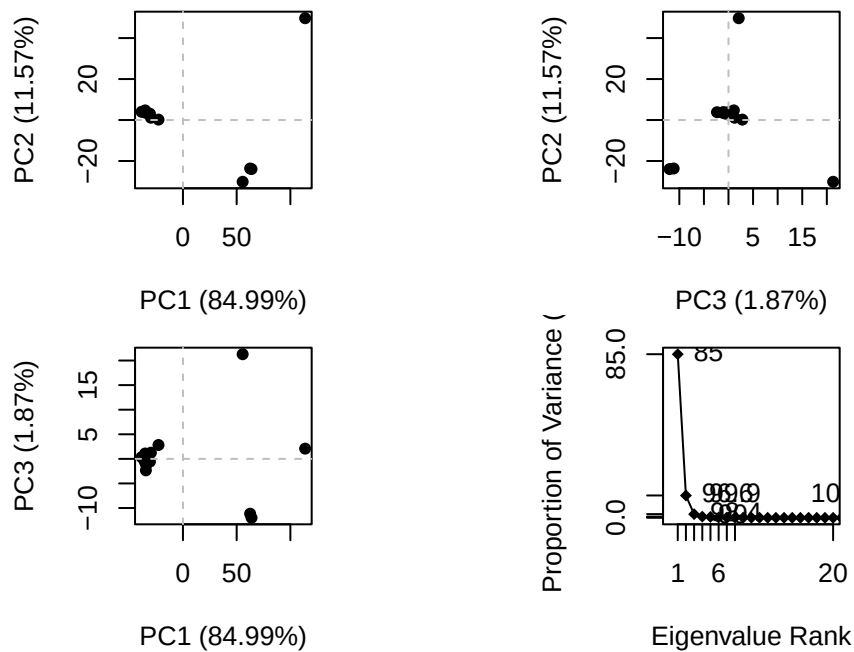
```
pdb/seq: 3    name: pdbname/split_chain/6RZE_A.pdb
      PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 4    name: pdbname/split_chain/3HPR_A.pdb
      PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 5    name: pdbname/split_chain/1E4V_A.pdb
pdb/seq: 6    name: pdbname/split_chain/5EJE_A.pdb
      PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 7    name: pdbname/split_chain/1E4Y_A.pdb
pdb/seq: 8    name: pdbname/split_chain/3X2S_A.pdb
pdb/seq: 9    name: pdbname/split_chain/6HAP_A.pdb
pdb/seq: 10   name: pdbname/split_chain/6HAM_A.pdb
      PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 11   name: pdbname/split_chain/4K46_A.pdb
      PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 12   name: pdbname/split_chain/3GMT_A.pdb
pdb/seq: 13   name: pdbname/split_chain/4PZL_A.pdb
```

```
library(bio3dview)
```

```
view.pdbname(pdbname)
```



```
pc.xray <- pca(pdbbs)
plot(pc.xray)
```

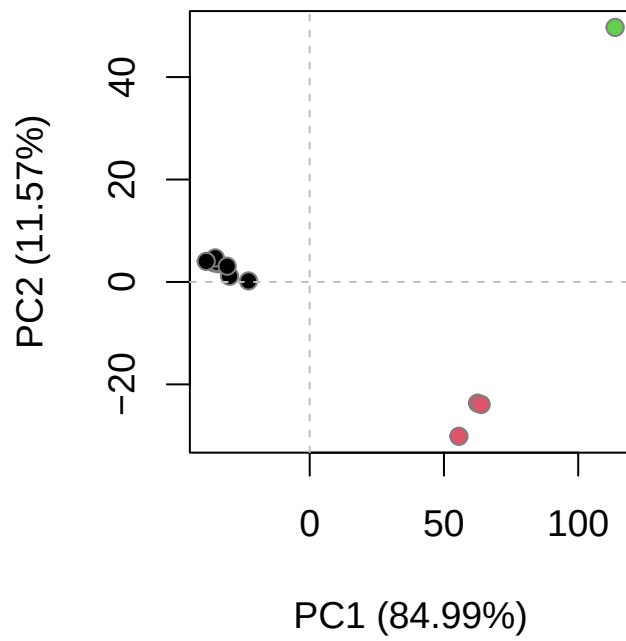


```
rd <- rmsd(pdbbs)
```

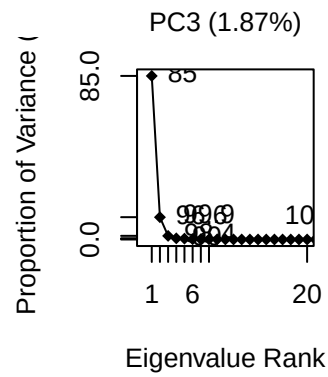
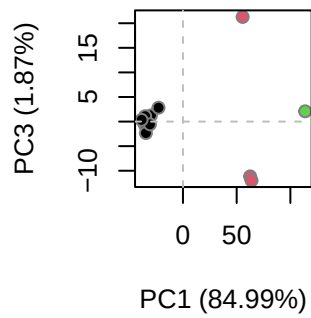
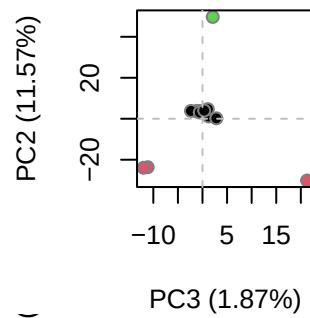
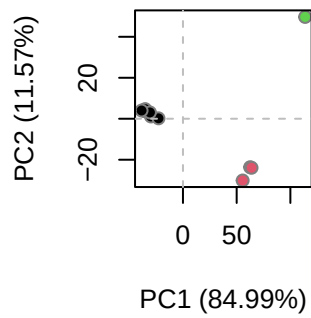
Warning in rmsd(pdbbs): No indices provided, using the 204 non NA positions

```
# Structure-based clustering
hc.rd <- hclust(dist(rd))
grps.rd <- cutree(hc.rd, k=3)
```

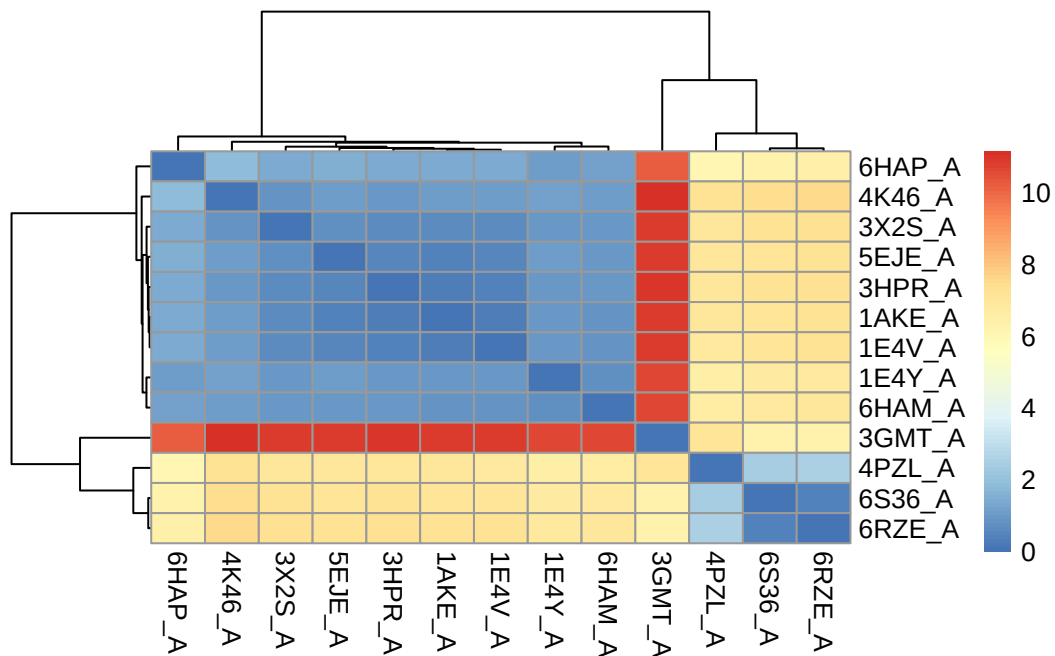
```
plot(pc.xray, 1:2, col="grey50", bg=grps.rd, pch=21, cex=1)
```



```
plot(pc.xray, col="grey50", bg=grps.rd, pch=21, cex=1)
```



```
library(pheatmap)
pheatmap(rd)
```



```
pc1 <- mktrj(pc.xray, pc=1, file="pc_1.pdb")
```

```
pdb <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```
view.pdb(pdb)
```

```
library(NGLViewer)
view.pca(pc.xray) |>
  setSpin()
```

